

Miles R. Meloni

Creative Technologist | GIS & Geospatial Visualization | Python | Data Analysis | NASA Experience | Frontend Development

CONTACTS

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SKILLS

Programming: Python (Pandas, Numpy, Matplotlib, Scipy), C, Java, JavaScript, R, Bash, CSS, HTML, Pug, LaTeX, C++, Haskell

GIS/Geospatial: ArcGIS Studio, ArcGIS Online, Google Earth Studio, GIS Workflows Spatial Analysis, Story Maps

Creative Skills: Writing, Editing (Video and Text), Office Suite, Adobe Suite (Photoshop, Illustrator, Premiere), Unity, Nvidia Omniverse, Google Web Suite

Development: Git, Gitlab, Github, VS Code, IntelliJ IDEA, Docker, Node.js, MongoDB, Terminal (Linux/Windows/Mac)

Research/Simulation: Finite Element Modeling, Autodesk Fusion, Quantum Mechanics, Computer Modeling, Virtual Reality

Interpersonal Skills: Communication, Conflict Resolution, Documentation, Teamwork, Adaptability, Creativity.

INTERNSHIP EXPERIENCE

NASA Jet Propulsion Laboratory Digital Twin Interoperability Researcher (May-Sept. 2025)

- Designed interoperable workflows to integrate cross-platform digital twin systems for use in future lunar missions.
- Utilized Generative Systems (Gemini, Claude) and cutting edge platforms (Unity, Unreal Engine, Nvidia Omniverse).
- Implemented Interpersonal Skills to pursue success in organizing multifaceted initiatives.

NASA Knowledge Management Intern, Langley Research Center (May-Sept. 2024)

- Directed, scripted, and animated three educational videos as NASA knowledge management resources.
- Presented work alongside senior NASA staff at Langley Research Center.
- Coordinated voice acting and video exporting and contributed to team meetings, product planning, and quality control.

VOLUNTEER EXPERIENCE

Student Assistantship, ESRI Developer Conference (Mar. 2026)

- Supported operations for the 2026 ESRI Developer Conference, serving approximately 2000 attendees.
- Engaged with GIS developers and gained a deep understanding of industry trends such as agentic workflows and interoperability and their applications within ArcGIS.

Environmental Education Internship, Cold Spring Harbor Fish Hatchery & Aquarium (Jun-Sept. 2023)

- Worked at a non-profit organization dedicated to educating students from kindergarten to 8th grade about the natural world. Engaged with students to foster their interest in nature and environmental stewardship.
- Took on a leadership position in educational programs and instruction of students.

RESEARCH

Experimental Physics Finite Element Simulation Researcher, LNGS and Bucknell University (Sept-May 2025/6)

- Conducted technical simulations on XENONnT detector using COMSOL to produce novel findings.
- Determined electrical field strength within detector using Python and Jupyter Notebooks
- Produced Honors Thesis on novel results and presented them to LNGS (Mar 2026).

Experimental Physics Finite Element Simulation Researcher, Bucknell University (Sept-Jan. 2023/5)

- Designed, constructed, and assembled parts for miniaturized dark matter detector.
- Analyzed anatomy of miniaturized detector to prove viability of internal electric fields using COMSOL.

Distributed Computing Algorithms Researcher, Bucknell University (Sept-May 2024/6)

- Studied asynchronous quorum systems used in distributed computing and analyzed algorithm efficiency.
- Utilized Rust code to experimentally determine viability of theoretical results.

Virtual Reality Visualization and 3D Modeling Designer, Bucknell University (Sept-Sept. 2022/4)

- Used virtual reality software to examine molecular structures and produce 3D models as education aids.
- Applied skills in virtual reality, Nanome, Mercury, Fusion 360, and Google Web Apps Suite

Brookhaven National Lab Student Intern/ Researcher (May-Sept. 2021)

- Implementation of Quantum Algorithms using Matrix Algebra
- Worked with Python coding and researched the structure of Python computers to demonstrate quantum supremacy

ADDITIONAL WORK

- Bucknell Tutor**, Teaching and Learning Center (Sept-May 2023/6)
- Led computer science and physics teaching lectures, taught coding knowledge in Python and data structures
 - Assisted students in debugging and troubleshooting code

EDUCATION

Bucknell University, Lewisburg, Pennsylvania. Dean's List, Tau Beta Pi member. (Graduation: May 2026)

Study Abroad: DIS Copenhagen (Jan-May 2025)

Major: Computer Science & Engineering

Minors: Physics, Mathematics, Environmental Studies

Presidential Fellowship Program Director

Bucknell Student Organizations: Science Fiction Club President, Creative Writing Club, Physics Society

AWARDS AND PUBLICATIONS

Bucknell's 30 Under 30 - Artistic Visionary- 2024

NASA's Shining Star Intern Award Nominee - 2024

Terra Firma- Serially published multimedia web novel on divinesubstances.com (art/creative writing portfolio)

Published in Confettihead (Bucknell Literary Magazine)